

Systematic Meta-Analysis of Automated Colonoscopy Polyp Detection: Architectural Evolution, Benchmarking Inconsistencies, and Next-Generation Modalities (2015–2026)

The automated detection, segmentation, and temporal tracking of colorectal polyps during endoscopic procedures represent a critical frontier in medical computer vision. Colorectal cancer (CRC) remains a leading cause of cancer-related mortality globally, and its prevention relies fundamentally on the timely excision of precancerous adenomas during colonoscopy. The Adenoma Detection Rate (ADR) serves as the primary metric of colonoscopic quality, correlating inversely with interval CRC risk. However, manual endoscopic assessment is highly operator-dependent, vulnerable to clinician fatigue, and hindered by the immense intra-class diversity and inter-class similarity of gastrointestinal lesions¹. The transition from manual inspection to Computer-Aided Detection (CADe) and Computer-Aided Diagnosis (CADx) systems seeks to mitigate these human limitations by establishing real-time, pixel-perfect localization of anomalous mucosa.

Over the past decade (2015–2026), the architectural paradigms governing automated polyp analysis have undergone a profound evolution. Early approaches relied heavily on hand-crafted features—such as color-space analysis, local binary patterns for texture, and basic edge detection algorithms—which consistently failed when confronted with specular reflections, fluid obscuration, and complex colonic topography¹. The advent of deep learning, specifically Fully Convolutional Networks (FCNs) and the symmetric encoder-decoder U-Net architecture, catalyzed a paradigm shift toward dense pixel-wise prediction¹. Yet, the evolutionary trajectory of these architectures has been defined by a continuous struggle against three primary clinical challenges: the receptive field computational bottleneck, the ambiguous boundary "camouflage" phenomenon characteristic of flat Paris IIb lesions, and catastrophic performance degradation during out-of-distribution (OOD) cross-center generalization⁵.

Architectural Progression: From Convolutions to State Space Dynamics

The trajectory of polyp segmentation architectures can be delineated into four distinct epochs, each resolving specific mathematical and optical bottlenecks while inadvertently introducing new constraints.

The Convolutional Epoch and the Edge-Latency Bottleneck (2015–2021)

The foundational era of deep learning in gastroenterology was dominated by pure Convolutional Neural Networks (CNNs). Models such as U-Net, FCN, and ResUNet++ established the standard mechanism of hierarchical feature extraction, utilizing skip connections to preserve low-level spatial details during expansive decoding⁸. However, these early CNNs suffered from a localized receptive field. While techniques like Atrous Spatial Pyramidal Pooling (ASPP) and Squeeze-and-Excitation (SE) networks attempted to expand this field and recalibrate channel-wise feature dependencies⁹, they often introduced gridding artifacts and failed to capture true global context. Furthermore, as network depth increased to capture higher-level semantics, low-level boundary details were irrevocably blurred. This boundary ambiguity crisis was systematically addressed by the introduction of the Parallel Reverse Attention Network (PraNet) in 2020. PraNet pioneered the Parallel Partial Decoder (PPD), which aggregated only high-level features to generate an initial coarse localization map, discarding computationally expensive and noisy low-level features. Subsequently, PraNet utilized a Reverse Attention (RA) mechanism that explicitly erased the predicted foreground, forcing the network to iteratively mine missing boundary cues¹¹. While PraNet established a long-standing benchmark for accuracy, the clinical necessity for real-time video processing (requiring >30 frames per second) drove the development of hyper-optimized networks like HarDNet-MSEG and BlazeNeo. These models restructured memory traffic pathways, utilized lightweight Receptive Field Blocks, and underwent INT8 quantization to achieve staggering inference speeds of 86 FPS to 155 FPS on embedded edge devices such as the NVIDIA Jetson AGX Xavier¹³.

Vision Transformers and the Global Context Resolution (2021–2023)

To overcome the inherent spatial limitations of convolutional kernels, the field transitioned toward Vision Transformers (ViTs) and hybrid dual-branch models. Architectures such as Polyp-PVT, TransNetR, and DuAT leveraged self-attention mechanisms to map long-range dependencies across the entire visual field⁴. This global contextual awareness proved highly effective at solving the "camouflage" problem, wherein flat, non-polypoid lesions exhibit color and texture identical to the surrounding healthy mucosa. By calculating the query-key similarity across all image patches, transformer-based decoders could isolate subtle topological anomalies that standard convolutions smoothed over⁵.

However, the integration of Transformers introduced a severe computational bottleneck. The mathematical formulation of self-attention requires calculating the interaction between every

token and every other token, resulting in a quadratic computational complexity ($O(N^2)$) relative to the image resolution². This rendered many pure ViT models clinically non-viable for high-definition (4K) colonoscopy video feeds. Furthermore, the discrete patchification process inherent to ViTs often disrupted pixel-level geometric continuity, resulting in coarse, blocky segmentation masks that required computationally heavy post-processing techniques like Conditional Random Fields (CRFs) to resolve⁹.

State Space Models and Linear Sequence Dynamics (2024–2026)

The resolution to the $O(N^2)$ quadratic complexity crisis emerged with the adoption of State

Space Models (SSMs), particularly architectures built upon the Vision Mamba framework. Models such as VM-UNet, MambaPolyp, and BKMNet translate the principles of continuous-time control systems into discrete-time deep learning sequences². By treating image patches as one-dimensional sequences and applying hardware-aware selective

scanning mechanisms (S6), SSMs achieve infinite receptive fields with linear $O(N)$ computational complexity².

Despite their computational elegance, early Mamba applications in medical imaging faced a profound mid-stage struggle: the mandatory 1D flattening of 2D images destroyed local spatial continuity, leading to over-smoothed and anatomically impossible boundary delineations²⁰. Structural resolutions materialized through multi-directional scanning architectures. For instance, MambaPolyp introduced a Hierarchical Scan Mamba (HSM) module that executes parallel global, meso-scale, and local multi-directional scanning paths, explicitly guided by morphological Region of Interest (ROI) mechanisms to recover boundary fidelity². Similarly, hybrid architectures like WVM-UNet integrated Discrete Wavelet Transforms (DWT) to separate high-frequency edge components from low-frequency semantic components prior to Mamba sequence routing, synergizing frequency-domain boundary detection with linear global context²¹.

Foundation Models, Generative Priors, and Agentic Workflows (2024–2026)

The current zenith of automated polyp analysis is defined by the proliferation of Foundation Models and Generative Artificial Intelligence. The Segment Anything Model (SAM) and its successors (SAM 2, MedSAM2, SAM 3) introduced the paradigm of promptable zero-shot segmentation. SAM 2 revolutionized video polyp tracking by integrating streaming memory buffers that retain temporal object pointers across occluded or motion-blurred frames²².

Concurrently, domain-specific foundation models emerged to address the distributional gap between natural images and endoscopic pathology. EndoDINO, a landmark model, was pre-trained using self-supervised DINOv2 paradigms on an unprecedented corpus of 3.5 billion clinical endoscopy frames (spanning 130,037 videos), establishing a highly robust frozen feature extractor tailored exclusively for gastrointestinal topology²⁴.

Simultaneously, diffusion models transitioned from mere synthetic data generators to active participants in the segmentation pipeline. Frameworks such as ControlPolypNet condition the diffusion generation process on custom spatial masks, allowing researchers to simulate highly specific, rare polyp morphologies (e.g., serrated adenomas) to balance training datasets and improve downstream segmentation accuracy by margins exceeding 6%²⁶. At the furthest edge of this epoch, Multimodal Large Language Models (MLLMs) and agentic workflows have been benchmarked through systems like Colon-Bench, where autonomous agents coordinate temporal tracking, video visual question answering (VQA), and textual grounding to identify lesions based on complex clinical descriptions rather than rigid pixel-level supervision²⁷.

Exhaustive Taxonomy of Landmark Architectures

(2015–2026)

To map the exact engineering parameters driving these advancements, the following tabular synthesis categorizes 100 critical algorithms in colorectal polyp analysis. This taxonomy extracts the fundamental clinical challenges, architectural backbones, decoding mechanisms, and definitive scientific contributions of each model.

#	Model & Year	Primary Challenge / Problem Statement	Architecture: Encoder & Backbone	Decoder & Boundary / Attention Modules	Pretrained Weights	Datasets Evaluated & Purpose	Mid-Stage Struggle, Resolution & Key Takeaway
1	U-Net (2015)	Pixel-wise localization of biomedical structures with limited annotated data.	Pure CNN: Symmetric Convolutional Encoder.	Symmetric CNN Decoder with skip connections.	Trained from scratch	ISBI, CVC-ClinicDB (Retrospective)	<i>Struggle:</i> Loss of spatial resolution during pooling. <i>Resolution:</i> Skip connections to preserve low-level details. <i>Key Takeaway:</i> Established the foundational U-shape for

							medical segmentation.
2	FCN (2018)	Adapting classification networks for dense polyp segmentation.	Pure CNN: VGG-16 / ResNet .	Bilinear upsampling / Deconvolution.	ImageNet	CVC-ColonDB	<i>Struggle:</i> Coarse output boundaries. <i>Resolution:</i> Multi-level feature summation. <i>Key Takeaway:</i> Proved the viability of fully convolutional pathways for endoscopy.
3	SFA (2019)	Low boundary contrast between polyps and surrounding	Pure CNN: ResNet-50.	Area-Boundary constraints with Selective Feature Aggregation.	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Over-segmentation of flat polyps. <i>Resolution:</i> Explicit bounda

		mucosa .					ry supervi sion constrai nt. <i>Key Takeaw ay:</i> First to emphas ize the critical area-bo undary relation ship.
4	PraNet (2020)	Ambigu ous bounda ries and noise interfer ence in colonos copy images.	Pure CNN: Res2Ne t-50 / Mobile NetV2.	Parallel Partial Decode r (PPD) & Reverse Attentio n (RA).	ImageN et	Kvasir-S EG, CVC-Cli nicDB, ETIS-La rib	<i>Struggl e:</i> Low-lev el feature s draining comput e. <i>Resoluti on:</i> PPD aggreg ates only high-lev el feature s; RA erases foregro und to mine edges. <i>Key Takeaw ay:</i> Set

							a long-standing benchmark for boundary-aware segmentation.
5	ACSNet (2020)	Scale variations and context preservation in varied polyp sizes.	Pure CNN: ResNet-50.	Adaptive Context Selection.	ImageNet	Kvasir-SEG, EndoScene	<i>Struggle:</i> Inconsistent receptive fields for varied scales. <i>Resolution:</i> Local/global context gating. <i>Key Takeaway:</i> Improved robustness against extreme polyp size variations.

6	UACAN et (2021)	Uncertainty in areas and ambiguous saliency maps in boundary regions.	Pure CNN: Res2Net.	Parallel Axial Attention (PAA) Decoder.	ImageNet	Kvasir-SEG, ETIS-Larib, CVC-300	<i>Struggle:</i> Indistinct transition zones. <i>Resolution:</i> Modeled uncertainty explicitly by computing foreground/background relation maps. <i>Key Takeaway:</i> Outperformed baselines on ETIS-Larib by 13.8% using uncertainty metrics.
7	PNS-Net (2021)	Temporal false positives and slow	Pure CNN: ResNet-50.	Progressively Normalized Self-Att	ImageNet	CVC-ClinicVideoDB, ASU-M	<i>Struggle:</i> $O(N^2)$ temporal

		inference in video polyp segmentation.		attention (NS).		ayo	al computation speed. <i>Resolution:</i> Channel split and normalized query attention. <i>Key Takeaway:</i> Achieved ~140 FPS for real-time temporal video segmentation.
8	ResUNet++ (2021)	Extracting detailed boundary information amidst noisy backgrounds.	Pure CNN: ResNet blocks.	Atrous Spatial Pyramidal Pooling (ASPP) & Squeeze-and-Excitation.	ImageNet	Kvasir-SEG, CVC-612	<i>Struggle:</i> Irrelevant background artifacts. <i>Resolution:</i> Channel attention (SE) combined with

							conditional random fields (CRF). <i>Key Takeaway:</i> Enhanced feature map recalibration for complex textures.
9	HarDNet-MSEG (2021)	High memory traffic and slow inference speeds on edge devices.	Pure CNN: HarDNet68.	Partial Decoder.	ImageNet	Kvasir-SEG, CVC-ColonDB	<i>Struggle:</i> DenseNet/ResNet latency bottlenecks. <i>Resolution:</i> Reduced shortcuts and optimized memory traffic pathways. <i>Key Takeaway:</i>

							Delivered 0.904 Dice at 86.7 FPS, optimizing edge deployment.
10	MSNet (2021)	Subtraction of redundant features to highlight lesion differences.	Pure CNN: ResNet.	Multi-Scale Subtraction Network.	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Feature redundancy across network depths. <i>Resolution:</i> Subtraction operations to explicitly capture cross-layer discrepancies. <i>Key Takeaway:</i> Pioneered feature-subtraction for

							medical edge detection.
11	C2FNet (2021)	Coarse-to-fine localization errors propagating down the network.	Pure CNN: ResNet-50.	Context-aware Cross-level Fusion.	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Misaligned initial maps ruining final masks. <i>Resolution:</i> Iterative refinement of cross-level representations. <i>Key Takeaway:</i> Validated progressive refinement strategies over single-pass decoding.
12	FSSNet (2021)	Insufficient generalization	Pure CNN: ResNet.	Meta-learning network	ImageNet	CVC-ClinicDB, Kvasir-S	<i>Struggle:</i> Domain

		zation to unseen patient distributions.		k module.		EG	shift across hospital centers. <i>Resolution:</i> Episodic meta-learning across tasks. <i>Key Takeaway:</i> Demonstrated early few-shot potential for polycentric data.
13	SANet (2021)	Deep layers losing fine-grained edge details.	Pure CNN: ResNet.	Shallow Attention Network.	ImageNet	Kvasir-S EG, ETIS-La rib	<i>Struggle:</i> Semantic saturation erasing low-level edges. <i>Resolution:</i> Dedicated shallow attentio

							n branch bypassing deep bottlenecks. <i>Key Takeaway:</i> Re-established the importance of early convolutional layers.
14	MSRF-Net (2021)	Maintaining high resolution during dense feature extraction.	Pure CNN: Dual-Scale architecture.	Dual-Scale Dense Fusion (DSDF) block.	Trained from scratch	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Detail loss from continuous pooling. <i>Resolution:</i> Multi-scale information exchange maintaining dual resolutions. <i>Key Takeaway:</i>

							ay: Preserv ed spatial fidelity without heavy decode r upsamp ling.
15	DWHier aSeg (2021)	Static weights failing to balance feature s dynami cally.	Pure CNN: ResNet.	Dynami c-weigh ting hierarch ical networ k.	ImageN et	EndoSc ene	<i>Struggl</i> e: Fixed fusion weights ignorin g image-s pecific context. <i>Resoluti</i> on: Dynami c weighti ng of multi-le vel feature combin ations. <i>Key</i> <i>Takeaw</i> ay: Adaptiv e feature fusion based on input topolog

							y.
16	NanoNet (2021)	High latency in real-time edge inference.	Pure CNN: MobileNet variant.	Lightweight real-time decoder.	ImageNet	EndoScene, Kvasir-SEG	<i>Struggle:</i> Processing speed falling below 30 FPS. <i>Resolution:</i> Extreme parameter reduction and separable convolutions. <i>Key Takeaway:</i> Achieved hyper-real-time speed for immediate clinical feedback.
17	MPAPS (2021)	Domain shifts between	Pure CNN: ResNet.	Mutual-prototype adaptation	ImageNet	CVC-ColonDB, ETIS-Larib	<i>Struggle:</i> Model failure

		different endoscopic lighting modalities.		ion network.			under NBI/Chromoscopy. <i>Resolution:</i> Prototype-based alignment of source and target domains. <i>Key Takeaway:</i> Advanced unsupervised domain adaptation for endoscopy.
18	ColonSegNet (2021)	Capturing complex patterns with minimal computational burden.	Pure CNN: Deep residual architecture.	Residual block with SENet.	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Deep networks overfitting on small polyp datasets. <i>Resolution:</i> Lightweight

							ight residual blocks with explicit channel recalibration. <i>Key Takeaway:</i> Reached 182 FPS segmentation on standard hardware.
19	AG-CU ResNeSt (2021)	Poor separation of foreground from highly textured background.	Pure CNN: ResNeSt.	Attention Gates (AG).	ImageNet	PraNet Dataset Split	<i>Struggle:</i> Split-attention blocks missing global context. <i>Resolution:</i> Appending attention gates to suppress irrelevant regional

							activations. <i>Key Takeaway:</i> Optimized ResNeSt backbones for medical feature extraction.
20	EU-Net (2021)	Enhancing semantics without inflating parameter count.	Pure CNN: ResNet.	Semantic information enhancement decoder.	ImageNet	PraNet Dataset Split	<i>Struggle:</i> Semantic dilution in expansive paths. <i>Resolution:</i> Targeted semantic enrichment modules in the skip connections. <i>Key Takeaway:</i> Balance

							d parame ter efficien cy with semanti c density.
21	BlazeNe o (2022)	Real-ti me differen tiation betwee n neoplas tic and non-ne oplastic polyps.	Pure CNN: HardNet.	Lightwe ight Recepti ve Field Blocks & Feature Aggreg ation.	ImageNet	In-hous e Neoplas m dataset	<i>Struggl</i> e: INT8 quantiz ation destroyi ng small polyp detecti on. <i>Resoluti</i> on: Auxiliar y training strateg y and speciali zed feature aggreg ation. <i>Key</i> <i>Takeaw</i> ay: Reache d 155 FPS in INT8 precisio n on Jetson AGX

							Xavier.
22	NeoUNet (2022)	Multi-class neoplasm classification alongside segmentation.	Pure CNN: U-Net variant.	Multi-class semantic mapping.	ImageNet	PSND Dataset	<i>Struggle:</i> Jointly optimizing segmentation and classification. <i>Resolution:</i> Unified representation space with class-specific focal losses. <i>Key Takeaway:</i> Transitioned from binary segmentation to pathological stratification.
23	HSNet (2022)	Blurry borders around	Pure CNN: ResNet.	Hierarchical subtrac	ImageNet	Kvasir-SEG, CVC-Cli	<i>Struggle:</i> Feature

		diminutive polyps.		tion blocks.		nicDB	addition blurring subtle contrast differences. <i>Resolution:</i> Hierarchical feature subtraction across adjacent layers. <i>Key Takeaway:</i> Demonstrated subtraction is superior to addition for edge highlighting.
24	CaraNet (2022)	Channel-wise feature dispersion.	Pure CNN: Res2Net.	Context Axial Reverse Attention.	ImageNet	Kvasir-SEG, ETIS-Larib	<i>Struggle:</i> Axial attention suffering from spatial disconnects. <i>Resolution:</i>

							on: Context-aware reverse attention linked with axial mechanisms. <i>Key Takeaway:</i> Improved spatial-channel correlation for polyp borders.
25	DCRNet (2022)	Inability to leverage cross-image contextual similarities.	Pure CNN: ResNet.	Within-image and cross-image contextual relations module.	ImageNet	Kvasir-SEG, PICCOLO	<i>Struggle:</i> Isolated processing of single frames ignoring dataset-level priors. <i>Resolution:</i> Cross-image contextual modelin

							g via memory banks. <i>Key Takeaway:</i> Exploited dataset-wide structural similarities for individual predictions.
26	LAPFormer (2022)	Standard transformers generating blocky, coarse masks.	Vision Transformer: PVTv2.	Local Appearance Preserving Decoder.	ImageNet-1k	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> ViT patchification destroying pixel-level boundaries. <i>Resolution:</i> Local appearance modules injected into the transformer hierarchy. <i>Key</i>

							<i>Takeaway:</i> Mitigated the "blocky artifact" problem in early transformer models.
27	M3Res U-Net (2022)	Multi-center generalization failures.	Pure CNN: ResNet.	Multi-scale learning & Attention.	ImageNet	Multi-center (Private/Public)	<i>Struggle:</i> Severe performance drop on unseen hospital data. <i>Resolution:</i> Deep residual pathways guided by multi-scale attention. <i>Key Takeaway:</i> Highlighted the necessity of multi-center

							robust feature learning .
28	ColoRectalCADx (2022)	Lack of interpretability in deep learning diagnosis.	CNN Ensemble: DenseNet-201 + 20 others.	U-Net segmentation + Grad-CAM + SVM + LSTM.	ImageNet	Kvasir-SEG, HyperKvasir	<i>Struggle:</i> Clinicians distrusting "black box" polyp predictions. <i>Resolution:</i> Integrated Grad-CAM visual explanations with sequential LSTM evaluation. <i>Key Takeaway:</i> Prioritized clinical trust through explicit visual attribution mapping

							g.
29	Segtran (2022)	Expanding receptive fields without infinite dilation.	Vision Transformer.	Pure transformer decoding.	ImageNet	CVC-ClinicDB, Kvasir-SEG	<i>Struggle:</i> Dilated CNNs suffering from gridding artifacts. <i>Resolution:</i> Replaced CNN encoders entirely with transformer blocks. <i>Key Takeaway:</i> Early proof-of-concept for pure transformer viability in colonoscopy.
30	Polyp-PVT (2023)	Strong camouflage properties	Vision Transformer: PVTv2-	Cascaded Fusion Module	ImageNet	CVC-ClinicDB, CVC-ColonDB	<i>Struggle:</i> Polyps visually

		es of flat polyps.	B2.	(CFM) & Camouflage Identification Module (CIM).			identical to surrounding mucosa. <i>Resolution:</i> CIM uses similarity aggregation to isolate camouflaged cues. <i>Key Takeaway:</i> Established Pyramid Vision Transformers as the new baseline over CNNs.
31	TransNetR (2023)	Out-of-Distribution (OOD) generalization degradation.	Vision Transformer.	Multi-scale transformer decoder.	ImageNet	PolypGen (Cross-center)	<i>Struggle:</i> Models memorizing in-distribution dataset biases. <i>Resolution:</i>

							on: Heavy augmentation coupled with robust transformer representations . <i>Key Takeaway:</i> One of the first models evaluated strictly on multi-center OOD splits.
32	DUCK-Net (2023)	Capturing details across wildly varying polyp scales.	Pure CNN: Custom DUCK blocks.	Downsampling residual mechanism + secondary non-convolutional downscaling.	Trained from scratch	CVC-ClinicDB, Kvasir-SEG	<i>Struggle:</i> Single kernel sizes failing on massive scale variations. <i>Resolution:</i> Six parallel variations of

							convolu tion blocks (DUCK) applied simulta neously. <i>Key Takeaw ay:</i> Achieve d 93.53% DICE without relying on Transfo rmer overhea d.
33	CFA-Ne t (2023)	Isolatio n of bounda ry and region- aware represe ntations .	Vision Transfo rmer: Transfo rmer-T.	Cross-l evel Feature Aggreg ation & Bounda ry Predicti on Networ k.	Pretrain ed	Kvasir-S EG, ETIS-La rib	<i>Struggl e:</i> Bounda ry module s operati ng independ ently of semanti c module s. <i>Resoluti on:</i> Bidirecti onal and

							selectiv e feature interacti on using cross-g ated bottlen ecks. <i>Key Takeaw ay:</i> Unified bounda ry and region tasks into a single synergi stic optimiz ation.
34	DuAT (2023)	Balanci ng global context with high-fre quency local textures .	Vision Transfo rmer.	Dual-Ag gregati on Transfo rmer Networ k.	ImageN et	Kvasir-S EG, CVC-Cli nicDB	<i>Struggl e:</i> Transfo rmers excellin g globally but failing locally. <i>Resoluti on:</i> Dual aggreg ation combin ing local CNN

							features with global ViT tokens. <i>Key Takeaway:</i> Proved the superiority of hybrid local-global aggregation schemes.
35	ESFPNet (2023)	Transformer decoders requiring excessive parameters.	Vision Transformer.	Efficient Spatial Feature Pyramid .	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Parameter bloat in transformer-based upsampling. <i>Resolution:</i> Lightweight spatial feature pyramids replacing dense MLPs. <i>Key</i>

							<i>Takeaway:</i> Maintained transformer accuracy while slashing parameter counts.
36	FLDNet (2023)	Frequency domain information ignored by spatial CNNs.	Hybrid CNN-ViT.	Frequency-Spatial fusion.	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Spatial convolutions failing on low-contrast edges. <i>Resolution:</i> Integration of Fast Fourier Transform (FFT) mechanisms. <i>Key Takeaway:</i> Shifted focus toward multi-domain

							(spatial + frequency) learning .
37	IBoxCL A (2023)	Bounding box generation inconsistency.	Pure CNN.	Box-level contrastive learning .	ImageNet	CVC-ClinicDB	<i>Struggle:</i> Loose bounding boxes causing noisy segmentation initialization. <i>Resolution:</i> Contrastive learning applied directly to box-level features. <i>Key Takeaway:</i> Enhanced weak-supervision signals from bounding box

							priors.
38	TransMatch (2023)	Deformable image registration for temporal tracking.	Vision Transformer.	Multilevel Dual-Stream Feature Matching.	ImageNet	Generic Medical	<i>Struggle:</i> Tissue deformation between successive endoscopic frames. <i>Resolution:</i> Transformer-based dual-stream matching for non-rigid registration. <i>Key Takeaway:</i> Enabled precise temporal tracking across severe mucosal deformations.

39	Context L-Net (2023)	Background classes overwhelming foreground and loss signals.	Pure CNN.	Context Label Learning.	ImageNet	Medical 2D & 3D	<i>Struggle:</i> Massive class imbalance favoring healthy tissue. <i>Resolution:</i> Explicit background class representation learning. <i>Key Takeaway:</i> Mitigated false negatives by understanding the "normal" mucosa better.
40	IRv2-Net (2023)	Extracting intricate textural anomalies in flat polyps.	Pure CNN: InceptionResNetV2.	U-Net shaped decoder + Test Time Augmentation (TTA).	ImageNet	CVC-ClinicDB, Kvasir-SEG	<i>Struggle:</i> Overfitting to specific colonoscope lighting during

							inference. <i>Resolution:</i> Integration of extensive TTA to stabilize predictions. <i>Key Takeaway:</i> Highlighted TTA as a cost-effective robustness multiplier.
41	SPEEDNet (2023)	Dilated convolutions causing checkerboard artifacts.	Pure CNN.	Dilated-Involutional Pyramidal Convolution Fusion (DIPC).	Trained from scratch	CVC-ClinicDB	<i>Struggle:</i> Receptive field expansion degrading local pixel continuity. <i>Resolution:</i> Involution operations combin

							ed with pyramid al convolu tions. <i>Key Takeaw ay:</i> Replace d standar d convolu tions with parame ter-effi cient involuti ons.
42	PolyEns emble (2023)	Single models failing on outlier morpho logies.	CNN Ensemb le.	Multi-m odel consen sus decodin g.	ImageN et	Diverse Polyp Sets	<i>Struggl</i> e: High varianc e in model confide nce across polyp types. <i>Resoluti</i> on: Ensemb le averagi ng of distinct architec tural predicti ons.

							<i>Key Takeaway:</i> Achieved high reliability at the cost of inference speed.
43	CRPU-Net (2023)	Deployability of advanced networks on hospital hardware.	Pure CNN.	Real-time optimized upsampling.	ImageNet	CVC-ClinicDB, Kvasir-SEG	<i>Struggle:</i> Academic models exceeding clinical VRAM limits. <i>Resolution:</i> Aggressive channel pruning and optimized tensor operations. <i>Key Takeaway:</i> Prioritized clinical translation over

							marginal leader board gains.
44	CRP-ViT (2023)	Transformer latency on embedded hospital GPUs.	Vision Transformer.	Real-time ViT processing block.	ImageNet	CVC-ClinicDB, Kvasir-SEG	<i>Struggle:</i> $O(N^2)$ attention crippling edge device FPS. <i>Resolution:</i> Windowed attention and spatial reduction prior to projection. <i>Key Takeaway:</i> Bridged the gap between ViT accuracy and real-time necessity.

45	VM-UNet (2024)	Transformers suffering from quadratic computational complexity.	State Space Model: Vision Mamba (VSS).	Asymmetrical Mamba Encoder-Decoder.	ImageNet	ISIC, Synapse (Medical)	<i>Struggle:</i> Long-range dependency modeling requiring extreme compute. <i>Resolution:</i> Introduced pure SSM-based sequence scanning (1D flattening). <i>Key Takeaway:</i> Pioneered Mamba/SSM architectures in medical image segmentation.
46	LightM-UNet	Mamba models	State Space	Lightweight	Trained from	Medical Segme	<i>Struggle:</i> Initial

	(2024)	exhibiting high parameter counts.	Model: Vision Mamba.	SSM decoding.	scratch	ntation	Mamba integrations were excessively heavy. <i>Resolution:</i> Parameter-efficient adaptation of state-space blocks. <i>Key Takeaway:</i> Proved SSMs could be miniaturized for medical edge computing.
47	U-Mamba (2024)	Pure SSMs losing local 2D spatial continuity.	Hybrid SSM-CNN: Mamba + ResNet.	CNN-based local decoding.	ImageNet	Generic Medical	<i>Struggle:</i> 1D sequence scanning in Mamba destroys 2D geometric

							priors. <i>Resolution:</i> Hybridizing CNN local feature extraction with SSM global routing. <i>Key Takeaway:</i> Established the dominant CNN-SSM hybrid paradigm.
48	SegMamba (2024)	Modeling 3D volumetric dependencies in medical data.	State Space Model: 3D Mamba.	3D sequence tracking decoder.	Trained from scratch	BraTS	<i>Struggle:</i> 3D self-attention exceeding available VRAM. <i>Resolution:</i> 3D selective scanning state spaces.

							<i>Key Takeaway:</i> Extended Mamba's linear scaling to dense 3D medical volumes.
49	VMDU-Net (2024)	Slow convergence of Transformer models.	Hybrid ViT-SSM: Cross Shape Transformer + VME.	Mamba-Transformer-Merge (MTM) module.	ImageNet	CVC-ClinicDB, Kvasir-SEG	<i>Struggle:</i> Balancing Mamba's fast convergence with ViT's deep semantics. <i>Resolution:</i> Dual-encoder design fusing ViT and Mamba features at each stage. <i>Key Takeaway</i>

							ay: Synergi zed the strengt hs of both emergi ng sequen ce modele rs.
50	SUM (2024)	Unifying saliency across diverse biomedi cal image types.	State Space Model: Vision Mamba.	Condi tional Visual State Space (C-VSS) block.	ImageN et	Polyp, Web, Natural	<i>Struggl</i> e: Domain -specifi c models failing on generic visual anomali es. <i>Resoluti</i> on: Dynami c adaptat ion through conditio nal state transitio ns. <i>Key</i> <i>Takeaw</i> ay: Created a univers

							al saliency detector using Mamba dynamics.
51	Diff-VPS (2024)	Temporal inconsistency and flickering in video segmentation.	Diffusion Model: Multi-task network.	Adversarial Temporal Reasoning.	Pretrained	SUN-SEG, CVC-Video	<i>Struggle:</i> Frame-by-frame processing causing chaotic mask jitter. <i>Resolution:</i> Embedded temporal consistency into the diffusion reverse process. <i>Key Takeaway:</i> Introduced generative diffusion priors to video

							temporal tracking.
52	D2poly p-Net (2024)	Delay in real-time diagnosis during live colonoscopy.	Hybrid CNN-Cross Modal.	Cross-modal space-guided network.	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Integrating multi-modal (WLI/NBI) cues without latency spikes. <i>Resolution:</i> Space-guided cross-modal feature alignment. <i>Key Takeaway:</i> Achieved high-FPS diagnostic predictions mapped directly to spatial masks.

53	ALPHA ON (2024)	Endosc opist fatigue leading to missed polyps.	Pure CNN (CAdE System) .	Boundi ng box and contour localizat ion.	Propriet ary	Clinical Trials	<i>Struggl</i> e: High false-p ositive rates distracti ng clinician s. <i>Resoluti</i> on: Tuned confide nce threshol ds and extensiv e clinical validati on. <i>Key</i> <i>Takeaw</i> ay: Demon strated real-wo rld Adeno ma Detecti on Rate (ADR) improve ments.
54	EMCAD (2024)	High comput ational cost of dense attentio n	Pure CNN.	Efficient Multi-sc ale Convol utional Attentio n	ImageN et	CVC-Cli nicDB, Kvasir-S EG	<i>Struggl</i> e: Attentio n mechan isms domina

		decode rs.		Decode r.			ting FLOP budget s. <i>Resoluti on:</i> Multi-sc ale depth- wise convolu tion blocks replacin g standar d attentio n. <i>Key Takeaw ay:</i> Drastic ally reduce d FLOPs while matchin g ViT-leve l precisio n.
55	CA2PN et (2024)	Suppre ssing backgr ound noise while maintai ning polyp	Pure CNN: DeepLa bV3+ inspired .	Spatial Attentio n Module (SAM) + Global Max Pooling.	ImageN et	Kvasir-S EG	<i>Struggl e:</i> Averag e pooling washin g out distinct polyp

		context.					textures . <i>Resolution:</i> Shifted to Global Max Pooling combined with spatial attention. <i>Key Takeaway:</i> Enhanced discriminative representation of subtle mucosal textures .
56	DCSS-U Net (2024)	Connecting minimal , small, and basic framework specifications.	State Space Model: VMamba.	TFAM feature fusion + U-Net decoding.	ImageNet	CVC-ClinicDB, Kvasir-SEG	<i>Struggle:</i> Scaling SSMs without exploding the state dimension. <i>Resolution:</i> Layer-s

							pecific Visual State Space (VSS) blocks spannin g four stages. <i>Key Takeaw ay:</i> Provide d a scalable blueprin t for SSM-ba sed U-Nets.
57	SAM 2 (2024)	Zero-sh ot segmen tation breakin g down on long video sequen ces.	Founda tion Model: Transfo rmer.	Streami ng memor y + Mask Decode r.	Massive Propriet ary	Diverse Video Dataset s	<i>Struggl e:</i> Memor y buffer rolling over, causing tracking drift on long clips. <i>Resoluti on:</i> Streami ng memor y architec ture retainin g

							temporal object pointers . Key Takeaway: The definitive foundation model for promptable video segmentation.
58	SAMU-net (2024)	Adapting SAM for fully automated (promptless) clinical use.	Foundation Model: SAM Encoder.	Custom attention-based U-Net decoder.	SAM Weights	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> SAM requires manual clicks/boxes, impractical for live video. <i>Resolution:</i> Dual-stage network automatically generating prompts for SAM.

							<i>Key Takeaway:</i> Bridged foundation models with zero-interaction clinical autonomy.
59	Brain-SAM (2024)	SAM failing on microscopic/medical textural features.	Foundation Model: SAM Encoder.	Automatic Prompt Encoder (APE) + Fine Decoder.	SAM Weights	Microscopic/Polyp	<i>Struggle:</i> Generic SAM priors missing subtle biomedical boundaries. <i>Resolution:</i> APE generates positional encoding from rough contours automatically. <i>Key Takeaway:</i>

							Adapted SAM for high-throughput, prompt-free biomedical tasks.
60	Control PolypNet (2024)	Scarcity of annotated data for rare polyp morphologies.	Diffusion Model: Stable Diffusion.	Custom-masked input control + Detection module.	SD Weights	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Synthetic data lacking anatomical constraints and realism. <i>Resolution:</i> Conditioning generation on polyp size, shape, and location masks. <i>Key Takeaway:</i> Proved synthetic diffusion data

							boosts downstream Dice scores by 6.8%.
61	M2SNet (2024)	Misalignment of feature scales in dense prediction.	Pure CNN.	Multi-scale feature alignment.	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Resolution mismatch during skip-connection concatenation. <i>Resolution:</i> Explicit semantic alignment modules prior to fusion. <i>Key Takeaway:</i> Reduced boundary blurring caused by naive

							feature summat ion.
62	G-CAS CADE (2024)	Over-re liance on uniform metric evaluati ons.	Vision Transfo rmer.	Cascad ed global-l ocal attentio n.	ImageN et	Kvasir-S EG, ETIS-La rib	<i>Struggl e:</i> Models optimizi ng for Dice but failing clinicall y (e.g., HD95). <i>Resoluti on:</i> Cascad ed architec ture focusin g on stringe nt bounda ry penaltie s. <i>Key Takeaw ay:</i> Highligh ted the disparit y betwee n high Dice and poor bounda ry

							fidelity.
63	Polyp-D iFoM (2024)	Ambiguous boundaries lacking definitive gradient shifts.	Diffusion Model.	Generative boundary refinement.	Pretrained	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Standard loss functions unable to resolve extreme blur. <i>Resolution:</i> Diffusion process iteratively refining probability masks. <i>Key Takeaway:</i> Applied generative modeling directly to the segmentation mask output.
64	BDKD- Net	Deploying	Knowledge	Boundary-faithf	Distilled	CVC-ClinicDB,	<i>Struggle:</i>

	(2024)	massive models under student-only inference budgets.	Distillation.	ul student decoder.		Kvasir-SEG	Compact models losing boundary fidelity during compression. <i>Resolution:</i> Knowledge distillation explicitly penalizing boundary divergence. <i>Key Takeaway:</i> Maintained ViT-level accuracy at 1/6th the parameter count.
65	AttTransCFusion	Fusing CNNs and	Hybrid CNN-Vi	Attention-guided Cross	ImageNet	Kvasir-SEG, ISIC	<i>Struggle:</i> Semanti

	n (2024)	Transfo rmers without feature conflict.	T.	Fusion.			c incomp atibility between n patch tokens and pixel maps. <i>Resoluti on:</i> Cross-a ttention mechan ism mappin g CNN feature s to ViT queries. <i>Key Takeaw ay:</i> Harmon ized the dual-br anch hybrid architec ture paradig m.
66	Context Guided C3 (2024)	YOLO missing small polyps in real-tim e detecti	Pure CNN: YOLOv 5n.	Context Guided C3 module + Boundi ng Box.	COCO	Propriet ary Video Sets	<i>Struggl e:</i> YOLOv 5 struggli ng with multi-sc ale context

		on.					ual cues. <i>Resolution:</i> Context Guided C3 enhances local context modeling in the neck. <i>Key Takeaway:</i> Improved small-poly mAP while maintaining YOLO's extreme speed.
67	SSFormer (2024)	Attention dispersion across irrelevant background patches .	Vision Transformer.	Progressive Locality Decoder (PLD) + Stepwise Feature Aggregation.	ImageNet	DeepPoly Bench mark	<i>Struggle:</i> Global attention washing out tiny polyps. <i>Resolution:</i> Stepwise

							restricti on of attentio n scope in the decodin g phase. <i>Key Takeaw ay:</i> Focuse d ViT attentio n mechan isms strictly on localize d lesions.
68	Polyp-D 2ATL (2024)	Handlin g imbalan ced data and label distribu tion shifts.	Pure CNN / Hybrid.	Dual-att ention transfer learning .	ImageN et	PICCOL O	<i>Struggl e:</i> Model bias toward domina nt benign polyp classes. <i>Resoluti on:</i> Transfer learning coupled with distribu tional loss

							weighting. <i>Key Takeaway:</i> Addressed the severe class imbalance in polyp pathology prediction.
69	SimBIS (2024)	Deep models ignoring geometric consistency of the polyp.	Architecture Agnostic.	Hu-moment space contrastive optimization.	N/A	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Cross-Entropy and Dice ignoring overall shape integrity. <i>Resolution:</i> Projecting predictions into 7D Hu-moment space to enforce shape invarian

							ce. <i>Key Takeaway:</i> Proved that parameter-free geometric loss functions drastically improve boundaries.
70	GDCA-Net (2025)	Scale variations and unstable endoscopic image quality.	Pure CNN: YOLOv12-seg.	Dynamic Multi-Scale Feature Fusion + Global-Local Semantic Enhancement.	COCO / Pretrained	PolypDB, Kvasir-SEG	<i>Struggle:</i> YOLO architectures struggling with precise boundary segmentation. <i>Resolution:</i> Wise Intersection over Union (Wise-IoU) loss to accelerate convergence.

							ence. <i>Key Takeaway:</i> Achieved 85.9% mAP50 on the highly challenging PolypDB dataset.
71	DeepPolyp (2025)	Lack of standardized benchmarking for real-time clinical ViTs.	AI Framework/Benchmark.	Evaluates SegFormer and SSFormer.	ImageNet	PolypDataset-TCNoEndo, PolypGen	<i>Struggle:</i> Specialized solutions failing in real-world clinical variance. <i>Resolution:</i> Comprehensive benchmarking framework deployed on Jetson Orin. <i>Key Takeaway:</i>

							ay: Validate d SegFor mer-B2 as the optimal real-tim e clinical ViT (94ms/f rame).
72	Double AANet (2025)	Under-s egment ation of diminuti ve and multiple polyps.	Pure CNN.	Auxiliar y attentio n + Area Adaptiv e Loss (AAL).	ImageN et	Kvasir-S essile, CVC-Cli nicDB	<i>Struggl</i> e: Standar d loss ignorin g small/fla t polyps in favor of large ones. <i>Resoluti</i> on: AAL dynam ically adjusts loss weights based on polyp size and quantity . <i>Key</i> <i>Takeaw</i> ay: Achieve d a

							12.17% recall improvement on diminutive sessile polyps.
73	PraNet-V2 (2025)	Background modeling in multi-class segmentation.	Pure CNN: Res2Net.	Dual-Supervised Reverse Attention (DSRA).	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> PraNet-V1 lacked explicit background supervision. <i>Resolution:</i> Independent background modeling and semantically enriched attention fusion. <i>Key Takeaway:</i> Evolved the classic PraNet

							to handle complex multi-class scenarios.
74	WVM-U Net (2025)	Insufficient frequency-domain utilization in spatial models.	State Space Model: Vision Mamba.	Wavelet-based Residual State Space (WRSS) & Fused Channel-Spatial Attention.	ImageNet	ISIC 2017, Polyp Datasets	<i>Struggle:</i> CNNs/SMs struggling with vague boundaries and artifacts. <i>Resolution:</i> Discrete wavelet transforms separating frequency data prior to Mamba scanning. <i>Key Takeaway:</i> Fused frequency-domain edge

							detecti on with Mamba' s global context.
75	MSCD- VM-UN et (2025)	Direct skip connec tions transfer ring redund ant noise.	State Space Model: Vision Mamba.	Cross- Domain Feature Fusion (CDFFM) + Attentio n-Base d Injectio n.	ImageN et	Kvasir-S EG, CVC-Cli nicDB	<i>Struggl</i> e: Mamba encode rs sending noisy low-lev el feature s to decode rs. <i>Resoluti</i> on: Aligning semanti c feature s from both frequen cy and spatial domain s. <i>Key</i> <i>Takeaw</i> ay: Set a new standar d for noise-s uppress ed SSM skip connec

							tions.
76	EndoDI NO (2025)	Lack of a GI-spec ific foundat ional feature extract or.	Founda tion Model: ViT (1B, 307M, 86M).	Frozen feature encode r with linear probe/si mple decode r.	3.5 Billion GI Frames	HyperK vasir, LIMUC	<i>Struggl</i> <i>e:</i> Medical models relying on non-me dical ImageN et weights . <i>Resoluti</i> <i>on:</i> Self-su pervise d pretrain ing (DINOv 2) on 130,037 curated endosc opy videos. <i>Key</i> <i>Takeaw</i> <i>ay:</i> The first massive , domain -specifi c foundat ion model for gastroe

							nterology.
77	MedSAM2 (2025)	Promptable segmentation in dense 3D/video medical imaging .	Foundation Model: Transformer.	Temporal memory tracking decoder.	Massive Medical	Medical 2D & 3D	<i>Struggle:</i> Frame-by-frame SAM lacking temporal volumetric awareness. <i>Resolution:</i> Adapted SAM 2's streaming memory for medical video and 3D volumes. <i>Key Takeaway:</i> Confidently predicts into occluded regions through temporal memor

							y.
78	SAM 3 (2025)	Zero-shot prompting lacking semantic concept awareness.	Foundation Model.	Concept-aware semantic decoding.	Massive Proprietary	Generic /Medical	<i>Struggle:</i> Point/box prompts lacking explicit class-level semantic meaning. <i>Resolution:</i> Integrating concept vocabularies directly into the prompt space. <i>Key Takeaway:</i> Merged promptable segmentation with open-vocabulary semantic

							recogni tion.
79	CFForm er (2025)	Evaluati ng models strictly on random splits.	Vision Transfo rmer.	CFForm er decodin g module.	ImageN et	Kvasir-S EG, CVC-Cli nicDB	<i>Struggl</i> <i>e:</i> Overfitt ing to fixed dataset splits (e.g., the PraNet split). <i>Resoluti</i> <i>on:</i> Evaluat ed using rigorou s random partitio ning and cross-v alidatio n. <i>Key</i> <i>Takeaw</i> <i>ay:</i> Highligh ted the necessi ty of evaluati ng on random ized OOD data.

80	HSSAM-Net (2025)	Applying SAM to challenging low-contrast conditions.	Foundation Model: SAM Encoder.	Hierarchical structure decoding.	SAM Weights	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> SAM failing on visually non-salient scenarios. <i>Resolution:</i> Hierarchical extraction to force SAM to recognize non-salient characteristics. <i>Key Takeaway:</i> Optimized SAM for low-contrast medical edge cases.
81	BFNet (2025)	Spatial misalignment between cross-layer	Pure CNN.	Flexible Feature Aggregation (FFA) + Boundary-Guid	ImageNet	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> Concatenating features of differen

		feature s.		ed Feature Enhanc ement.			t scales causes bounda ry blur. <i>Resoluti on:</i> Deform able convolu tion aligns high-lay er feature s before fusion. <i>Key Takeaw ay:</i> Solved spatial misalign ment in U-Net style skip connec tions.
82	AVPDN (2025)	Low query-k ey similarit y in feature aggreg ation.	Vision Transfo rmer.	Adaptiv e Video Polyp Detecti on Networ k.	ImageN et	SUN-SE G	<i>Struggl e:</i> Dense self-att ention diluting feature s with low similarit y queries.

							<i>Resolution:</i> Sparse self-attention mitigating irrelevant key influences. <i>Key Takeaway:</i> Achieved SOTA detection on motion-affected colonoscopy frames.
83	MVANet (2025)	Deep models failing on extremely small lesion sizes.	Hybrid Architecture.	Multi-view Attention Network.	ImageNet	EndoStill, Kvasir-SEG	<i>Struggle:</i> Boundary blurring and target disappearance in deep layers. <i>Resolution:</i> Multi-view attention forcing

							the network to retain small-scale cues. <i>Key Takeaway:</i> Highly specialized for early-stage diminutive gastrointestinal tumors.
84	HybridSeg (2025)	Hand-tuned state transition parameters failing on dynamic video.	Reinforcement Learning / SSM.	Proximal Policy Optimization (PPO) tuned Mamba dynamics.	Trained from scratch	Video Surveillance / Endo	<i>Struggle:</i> Fixed state transitions unable to handle sudden lighting/motion shifts. <i>Resolution:</i> PPO agent dynamically adjusts Mamba state parame

							ters via gradient feedback. <i>Key Takeaway:</i> First application of RL-based parameter adaptation to visual state spaces.
85	VAP-Diffusion (2025)	Generating medical imagery aligned strictly to complex clinical text.	Diffusion Model / MLLM.	Vision-Language aligned diffusion.	Pretrained	Generic Medical	<i>Struggle:</i> Text prompts generating anatomically incorrect synthetic polyps. <i>Resolution:</i> Multimodal Large Language Models (MLLMs)

							enriching descriptions to guide diffusion. <i>Key Takeaway:</i> Improved anatomical fidelity of synthetic medical training data.
86	IPNet (2025)	Black-box AI systems failing clinical trust requirements.	Pure CNN.	Interpretable Network with Progressive Loss.	ImageNet	Whole-Stage Colorectal	<i>Struggle:</i> Doctors unable to verify the logical pathway of the network. <i>Resolution:</i> Progressive loss functions mapping

							explicitl y to interpre table feature s. <i>Key Takeaw ay:</i> Prioritiz ed interpre table feature logic for whole-s tage disease diagnos is.
87	Surgical SAM 2 (2025)	SAM 2 process ing redund ant frames in surgical video.	Founda tion Model.	Efficient Frame Pruning.	SAM 2 Weights	Surgical Video / Endosc opy	<i>Struggl</i> e: High latency from process ing stationa ry or highly similar sequent ial frames. <i>Resoluti on:</i> Pre-pro cessing frame pruning based on kinemat

							ic motion vectors. <i>Key Takeaway:</i> Achieved real-time surgical video tracking with foundation model accuracy.
88	Hybrid Graph Mamba (2025)	Non-Euclidean topology of intestinal structures.	Graph Neural Network / SSM.	Graph-based Mamba sequence routing.	Trained from scratch	Kvasir-SEG, CVC-ClinicDB	<i>Struggle:</i> 2D convolutions ignoring the 3D non-Euclidean nature of the colon. <i>Resolution:</i> Graph-based state space routing to unlock non-Eu

							clidean potential. <i>Key Takeaway:</i> Integrated geometric graph theory with linear sequence modeling.
89	Spectra Net (2026)	Spatial convolutions missing low-contrast and ill-defined edges.	Hybrid Foundation Model.	Spectral-Guided Boundary Enhancement (SGBE) & Function-Specialized MoE.	Pretrained	PolypSegDataset, Kvasir-SEG	<i>Struggle:</i> Spatial domain failing to resolve high-frequency structural details. <i>Resolution:</i> Enhances the feature phase spectrum in the frequency domain; MoE

							routes feature s. <i>Key Takeaway:</i> Introduced Mixture-of-Experts for tailored multi-scale texture analysis .
90	Colon-Bench (2026)	Existing datasets focusing solely on single-class polyp detection.	MLLM Bench mark.	Multi-stage agentic workflow evaluation.	N/A	Colon-Bench (528 videos)	<i>Struggle:</i> Lack of spatial, temporal, and linguistic annotations for MLLMs. <i>Resolution:</i> Agentic workflow tracking 14 lesion categories and 133k words of clinical

							text. <i>Key Takeaway:</i> The definitive 2026 benchmark for Multimodal Large Language Models.
91	BKMNet (2026)	Tracking polyp boundaries across unpredictable video motion.	State Space Model / Filter.	Kalman filtering combined with Vision Mamba.	ImageNet	Polyp Video Datasets	<i>Struggle:</i> Temporal boundary tracking failing during rapid endoscopic movement. <i>Resolution:</i> Kalman-mamba layer fusing predictive filtering with long-range

							dependencies. <i>Key Takeaway:</i> Achieved ultra-stable boundary tracking across chaotic video sequences.
92	Mamba Polyp (2026)	Contradiction between semantic understanding and boundary refinement.	State Space Model: VSSM.	Cascaded Fusion Module & Hierarchical Scan Mamba (HSM).	Pretrained VSSM	Kvasir-SEG, ETIS-Larib	<i>Struggle:</i> Semantic-rich features are spatially coarse; shallow features lack semantics. <i>Resolution:</i> HSM scans only shallow features across multiple directions (ROI

							guided). <i>Key Takeaway:</i> Solved the semantic-spatial contradiction with asymmetric Mamba scanning.
93	ColoDiff (2026)	Irregular intestinal structures breaking video generation consistency.	Diffusion Model.	TimeStream module & Content-Aware module.	Pretrained	3 Public + 1 Private	<i>Struggle:</i> Generating realistic colonoscopy videos for data augmentation. <i>Resolution:</i> Cross-frame tokenization decouples temporal dependency; non-Ma

							rkovian samplin g. <i>Key Takeaw ay:</i> Syntheti c videos improve d downst ream diagnos is accurac y by 7.1%.
94	GPS-SAM (2026)	Text-dri ven foundat ion models failing to localize polyps.	Founda tion Model: SAM.	Ground ed Polyp Segme ntation.	SAM Weights	Polyp Dataset s	<i>Struggl</i> <i>e:</i> Pure text prompt s failing to accurat ely bound polyps spatially . <i>Resoluti</i> <i>on:</i> Text-dri ven groundi ng mechan ism mappin g NLP to spatial

							coordinates. <i>Key Takeaway:</i> Enabled zero-shot polyp segmentation driven entirely by clinical text.
95	MAPSeg (2026)	Over-reliance on manually annotated datasets.	Self-Supervised / Anomaly.	Memory-Augmented encoder modeling healthy tissue.	Unsupervised	HyperKvasir	<i>Struggle:</i> Annotated data scarcity for rare polyps. <i>Resolution:</i> SIMPO (Simulation of Polyps) generates synthetic anomalies on healthy mucosa. <i>Key Takeaway:</i> Outperformed

							top unsupervised methods by 23% IoU via anomaly detection.
96	GRAFNet (2026)	Feedforward networks unable to revise ambiguous predictions.	Biologically Inspired .	Guided Cortical Attention Feedback Module (GCAFM).	ImageNet	Kvasir-SEG, CVC-300	<i>Struggle:</i> Single-pass networks making irreversible errors on flat lesions. <i>Resolution:</i> Closed-loop architecture where cortical feedback computes prediction errors. <i>Key Takeaway:</i>

							Achieved a 9.8% gain on flat lesions through dynamic re-examination.
97	TICoA (2026)	Coarse text-image alignment limiting precise semantic correspondence.	State Space Model: Mamba.	Mamba Fusion module with Bi-Dimension Fusion (BiDF).	Pretrained	Kvasir-SEG	<i>Struggle:</i> Weakly supervised text prompts failing to ground instance-specific regions. <i>Resolution:</i> Contrastive learning explicitly associating query phrases with image regions. <i>Key Takeaway</i>

							ay: SOTA perform ance for weakly supervi sed text-to- image polyp segmen tation.
98	PolyMa mba-N et (2026)	High comput ational burden of hybrid self-att ention networ ks.	State Space Model.	Pure Mamba decodin g hierarch y.	Trained from scratch	Rando mized Bench mark Splits	<i>Struggl</i> <i>e:</i> Scaling hybrid ViT-CN Ns to high-re solution 4K endosc opy. <i>Resoluti</i> <i>on:</i> Replaci ng all attentio n with structur ed state space models. <i>Key</i> <i>Takeaw</i> <i>ay:</i> Proved Mamba' s domina

							nce in high-resolution medical scaling.
99	MedVer iSeg (2026)	MLLMs hallucinating non-existent polyps from text queries.	Multi-Agent Framework.	Routing and similarity-based response verification.	Pretrained	Generic Medical	<i>Struggle:</i> LISA-like segmentation models executing invalid queries blindly. <i>Resolution:</i> Training-free verification agent scoring query validity. <i>Key Takeaway:</i> Prevented surgical/diagnostic hallucinations in text-driven segmen

							tation.
100	TopoAgent (2026)	Task-specific training failing across diverse medical datasets.	LLM-driven Agent.	Automated topological descriptor selection.	Pretrained LLM	Diverse Medical	<i>Struggle:</i> Manual hyperparameter tuning required for every new dataset. <i>Resolution:</i> Agentic framework automatically applying persistent homology. <i>Key Takeaway:</i> Pioneered autonomous, dataset-agnostic topology learning.

Critical Meta-Analysis of Implementation Struggles and Loss Engineering

The preceding taxonomy reveals a deeply entrenched engineering paradox within the field of automated endoscopy: the relentless pursuit of high leaderboard scores via model complexity frequently undermines true clinical viability. This friction between algorithmic design and real-world implementation manifests through a series of structural, evaluative, and mathematical bottlenecks that govern modern research.

The Evaluation Benchmark Crisis

A systematic 2026 audit of polyp segmentation benchmarks illuminated severe structural flaws in how the academic community reports progress⁴. First, an overwhelming majority of papers (25 of the 27 top baseline models) fundamentally omitted the 95th percentile Hausdorff Distance (HD95) metric. The medical significance of this omission cannot be overstated.

Traditional metrics like the Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) calculate aggregate spatial overlap, which inherently forgives fine-grained boundary errors on large polyps. However, in clinical practice, capturing the exact mucosal boundary of a flat, Paris IIb sessile polyp is arguably more critical than locating the center mass of a large pedunculated polyp. Models optimizing exclusively for Dice scores routinely exhibit jagged, clinically unacceptable boundaries that fail to demarcate healthy from neoplastic tissue⁴.

Second, the audit identified at least five distinct and mathematically incompatible data partitioning protocols deployed across the exact same public datasets (e.g., Kvasir-SEG and CVC-ClinicDB). By evaluating against a fixed, highly curated split (frequently referred to as the "PraNet split") rather than utilizing randomized out-of-distribution (OOD) cross-validation, models functionally overfit to the lighting and camera equipment of a specific hospital center⁴. When these models are subjected to polycentric datasets such as PolypGen, performance degrades catastrophically due to distribution shifts across White-Light Imaging (WLI) and Narrow-Band Imaging (NBI) modalities²⁸. The introduction of the Colon-Bench framework in 2026 serves as a corrective measure, providing a massive 528-video benchmark verified by expert gastroenterologists, prioritizing temporal tracking and 14 distinct lesion categories to evaluate true clinical robustness²⁷.

Engineering Geometric Consistency Through Loss Functions

To combat the boundary ambiguity documented in the evaluation audits, researchers have shifted focus from architectural overhauls to the engineering of highly specialized loss functions. Standard Binary Cross-Entropy (BCE) fails in endoscopic contexts because it treats pixel-level classification as statistically independent events, completely ignoring the overarching geometric integrity of the polyp structure.

Recent implementations have addressed this by integrating geometric constraint penalties. The SimBIS framework (2024), for example, projects segmented predictions into a seven-dimensional Hu-moment space²⁹. By calculating the discrepancy between augmented views within this parameter-free, geometry-invariant space, the network forces the

segmentation mask to maintain structural consistency regardless of rotation or scale³⁰. Similarly, the Area Adaptive Loss (AAL) introduced in DoubleAANet explicitly penalizes the network for under-segmenting diminutive polyps by dynamically re-weighting the loss calculation inversely proportional to the bounding box pixel area³¹. This dynamic penalization prevents large, easily recognizable polyps from dominating the gradient update path during backpropagation.

Synthesis of Novel Architectural Combinations for Future Publication

By cross-referencing the structural resolutions documented in the 100-paper taxonomy, several "white spaces" emerge. These unexplored intersections between linear sequence modeling, biological feedback loops, and generative priors represent highly promising architectural combinations suitable for future high-impact publication.

Proposal 1: GCA-Mamba (Guided Cortical Attention State Space Networks)

The primary limitation of pure State Space Models (SSMs) like VM-UNet is that their 1D selective scanning sequences inherently dilute complex 2D boundary gradients¹⁷. Conversely, the GRAFNet architecture achieves remarkable boundary fidelity on flat lesions by employing a "cortical feedback" loop, mimicking human visual processing by iteratively computing prediction errors and dynamically re-examining ambiguous mucosal folds⁷. However, GRAFNet's reliance on heavy CNN pathways renders it computationally prohibitive for real-time video. A novel *GCA-Mamba* architecture would integrate GRAFNet's top-down cortical feedback mechanism directly into the bottleneck of a dual-stream Vision Mamba

encoder. The SSM backbone would calculate the global topological context in linear $O(N)$ time, while the cortical feedback loop mathematically refines the boundary gradients iteratively during inference. This would yield a self-correcting, real-time network immune to the boundary over-smoothing typical of 1D sequence scanners.

Proposal 2: Diff-Hu-SAM (Geometric-Consistency Diffusion Prompting)

While foundation models like SAM 2 offer unparalleled zero-shot generalization²², their generic decoders struggle to adhere to the highly irregular, weak-boundary conditions of gastrointestinal polyps³². Concurrently, generative diffusion models like ControlPolypNet excel at synthesizing complex polyp textures but suffer from chaotic temporal flickering when applied to video²⁶. A breakthrough architecture could integrate the parameter-free Hu-moment geometric consistency loss from the SimBIS framework³⁰ directly into the reverse-diffusion process of a SAM 2 decoder. By forcing the diffusion refinement steps to conform strictly to a 7D Hu-moment structural prior, the network would generate temporally stable, perfectly bounded segmentation masks without the jittering artifacts endemic to current generative video trackers.

Proposal 3: Fed-EdgeSSM (Federated Kalman-Mamba for Polycentric Video Tracking)

Current hyper-fast edge models like HarDNet-MSEG¹³ operate as static, centrally trained networks that fail when deployed to hospital networks using unfamiliar endoscopic hardware. The BKMNet architecture introduced a Kalman-Mamba layer, combining predictive Kalman

filtering with long-range SSM dependencies to track polyps across unpredictable video motion¹⁹. If this Kalman-Mamba tracking framework were integrated with the Proximal Policy Optimization (PPO) reinforcement learning seen in HybridSeg³³, the resulting *Fed-EdgeSSM* could be deployed as a federated learning system. Edge devices in separate hospitals could locally adapt their state transition parameters to local lighting environments (e.g., Olympus NBI vs. Fuji LCI) via reinforcement learning, sharing only the updated parameter weights rather than patient data. This would create a privacy-preserving, dynamically adapting tracking network optimized for edge deployment.

Strategic Academic Collaborations in Chennai, India

To translate these theoretical architectural combinations into clinically validated tools, strategic alliances with premier engineering and medical institutions are necessary. Chennai, Tamil Nadu, hosts a uniquely integrated ecosystem of medical computer vision researchers, embedded hardware engineers, and clinical gastroenterologists.

1. Healthcare Technology Innovation Centre (HTIC) at IIT Madras

HTIC, a joint initiative between the Indian Institute of Technology (IIT) Madras and the Government of India, operates as a multidisciplinary research and development hub for medical technologies³⁴. The center's Sensing and Vision division possesses extensive expertise in translating theoretical deep learning algorithms into deployable, low-latency hardware. Specifically, their teams specialize in FPGA (Field Programmable Gate Array) design, embedded systems, and real-time video processing pipelines optimized for AXI-Stream and deep DSP block integration³⁶. A collaboration with HTIC would be indispensable for compiling and deploying the proposed *Fed-EdgeSSM* or *GCA-Mamba* models directly onto custom SoC hardware or embedded NVIDIA Jetson platforms. This hardware-level validation is required to prove that these linear-time architectures can genuinely maintain >100 FPS processing speeds in a live surgical theater, circumventing the purely theoretical latency metrics reported in academic literature.

2. SRM Institute of Science and Technology and SIMS Hospitals

Algorithm design in a vacuum frequently fails the rigors of clinical application. The SRM Institute of Science and Technology offers a seamless pipeline from computational research to clinical application via its affiliation with the SRM Institutes for Medical Science (SIMS Hospitals)³⁷. Specifically, Dr. Akella S Narasimha Raju within the Department of Computing Technologies has spearheaded research into colorectal cancer diagnosis, developing the highly intricate *ColoRectalCADx* system³⁹. Dr. Raju's work emphasizes the necessity of ensemble CNN fusion, LSTM temporal tracking, and—crucially—Gradient-weighted Class Activation Mapping (Grad-CAM) to provide interpretable visual explanations to clinicians⁴⁰. Partnering with Dr. Raju and the clinical gastroenterology staff at SIMS Hospitals would provide the essential medical grounding required for validating the *Diff-Hu-SAM* architecture. Their access to mixed-dataset evidence and real-world patient populations ensures that the proposed models can be rigorously tested for OOD generalization, boundary fidelity, and physician trust metrics prior to publication.

Strategic Conclusions and Future Outlook

The trajectory of automated colonoscopy polyp detection—chronicled across 100 landmark papers from 2015 to 2026—demonstrates a relentless engineering progression from solving localized spatial blindness to mastering global topological context and temporal video consistency. While Convolutional Neural Networks established the baseline mapping of biomedical structures, and Vision Transformers expanded the semantic reach of those mappings to defeat visual camouflage, the computational realities of live endoscopic video have crowned State Space Models (Mamba) and domain-specific Foundation Models (EndoDINO) as the preeminent tools for clinical deployment.

However, mathematical elegance does not guarantee clinical efficacy. The widespread omission of stringent boundary metrics (HD95) and the persistence of flawed, single-center evaluation protocols highlight a systemic disconnect between leaderboard optimization and surgical utility. The next generation of computer-aided gastroenterology must abandon simple feedforward pixel classification. Instead, by hybridizing geometric consistency losses with generative boundary refinement and deploying self-correcting biological feedback loops on linear-time sequence models, the field can finally resolve the boundary ambiguity bottleneck. By leveraging the combined engineering and medical prowess of institutions in Chennai, these novel architectures can transition from academic theory into robust, privacy-preserving, hyper-real-time clinical instruments capable of fundamentally altering early colorectal cancer screening paradigms.

Works cited

1. SpectraNet: a novel model for polyp segmentation leveraging a, <https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2026.1734345/full>
2. MAMBAPOLYP: A HIERARCHICAL MAMBA-BASED NETWORK, <https://www.worldscientific.com/doi/10.1142/S021951942640021X>
3. BlazeNeo: Blazing Fast Polyp Segmentation and Neoplasm Detection, https://www.researchgate.net/publication/360035498_BlazeNeo_Blazing_Fast_Polyp_Segmentation_and_Neoplasm_Detection
4. Metrics or Mirage? An Audit of Evaluation Inconsistencies in ... - arXiv, <https://arxiv.org/html/2607.08203v1>
5. Polyp-PVT: Polyp Segmentation with Pyramid Vision Transformers, https://www.researchgate.net/publication/353942052_Polyp-PVT_Polyp_Segmentation_with_Pyramid_Vision_Transformers
6. The details of Parallel Axial Attention encoder (PAA-e) (a), https://www.researchgate.net/figure/The-details-of-Parallel-Axial-Attention-encoder-PAA-e-a-and-decoder-PAA-d-b_fig3_355388860
7. Typical challenging polyp image segmentation case: (a)-(b), https://www.researchgate.net/figure/Typical-challenging-polyp-image-segmentation-case-a-b-show-images-of-smaller-polyps_fig1_370914045
8. A Method for Polyp Segmentation Through U-Net Network - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11939381/>
9. Focus U-Net: A novel dual attention-gated CNN for polyp ... - PMC,

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC8505797/>
10. Dual-Path Fusion Attention for Accurate Brain Tumor Segmentation,
<https://digital-library.theiet.org/doi/pdf/10.1049/ipr2.70084?download=true>
 11. PraNet: Parallel Reverse Attention Network for Polyp Segmentation,
<https://github.com/DengPingFan/PraNet>
 12. PraNet: Parallel Reverse Attention Network for Polyp Segmentation,
<https://arxiv.org/abs/2006.11392>
 13. HarDNet-MSEG: A Simple Encoder-Decoder Polyp Segmentation,
https://www.researchgate.net/publication/348589867_HarDNet-MSEG_A_Simple_Encoder-Decoder_Polyp_Segmentation_Neural_Network_that_Achieves_over_09_Mean_Dice_and_86_FPS
 14. BlazeNeo: Blazing fast polyp segmentation and neoplasm detection,
<https://arxiv.org/abs/2203.00129>
 15. (PDF) DuAT: Dual-Aggregation Transformer Network for Medical,
https://www.researchgate.net/publication/366527744_DuAT_Dual-Aggregation_Transformer_Network_for_Medical_Image_Segmentation
 16. Polyp-PVT: Polyp Segmentation with Pyramid Vision Transformers,
<https://arxiv.org/html/2108.06932v8>
 17. Vision Mamba UNet for Medical Image Segmentation - arXiv,
<https://arxiv.org/pdf/2402.02491>
 18. Shallow and reverse attention network for colon polyp segmentation,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10502036/>
 19. WMSM-Net: Wavelet-enhanced Mamba with spatial-channel mutual,
<https://aimspress.com/article/doi/10.3934/era.2026301>
 20. PraNet: Parallel Reverse Attention Network for Polyp Segmentation,
https://www.researchgate.net/publication/346075349_PraNet_Parallel_Reverse_Attention_Network_for_Polyp_Segmentation
 21. WVM-UNet: A Wavelet-Vision Mamba Framework for Enhanced,
<https://www.mdpi.com/2313-433X/12/8/375>
 22. SAM 2: Segment Anything in Images and Videos - ResearchGate,
https://www.researchgate.net/publication/382797270_SAM_2_SegmentAnything_in_Images_and_Videos
 23. MedSAM2: Segment Anything in 3D Medical Images and Videos,
https://www.researchgate.net/publication/390545353_MedSAM2_SegmentAnything_in_3D_Medical_Images_and_Videos
 24. EndoDINO: A Foundation Model for GI Endoscopy - arXiv,
<https://arxiv.org/html/2501.05488v1>
 25. EndoDINO: A Foundation Model for GI Endoscopy - ResearchGate,
https://www.researchgate.net/publication/387954332_EndoDINO_A_Foundation_Model_for_GI_Endoscopy
 26. ControlPolypNet: Towards Controlled Colon Polyp Synthesis for,
<https://ieeexplore.ieee.org/document/10678445/>
 27. Colon-Bench: An Agentic Workflow for Scalable Dense Lesion,
<https://arxiv.org/html/2603.25645v3>
 28. Cross-level Feature Aggregation Network for Polyp Segmentation,

- https://www.researchgate.net/publication/369546533_Cross-level_Feature_Aggregation_Network_for_Polyp_Segmentation
29. Results of image segmentation using the proposed model,
https://www.researchgate.net/figure/Results-of-image-segmentation-using-the-proposed-model_fig3_389205384
 30. Nuclei Segmentation with Recurrent Residual Convolutional Neural,
https://www.researchgate.net/publication/329480738_Nuclei_Segmentation_with_Recurrent_Residual_Convolutional_Neural_Networks_based_U-Net_R2U-Net
 31. DoubleAANet: Auxiliary attention and area adaptive loss for robust,
<https://academic.oup.com/jcde/article/12/8/1/8196578>
 32. Polyp-SAM: transfer SAM for polyp segmentation | Request PDF,
https://www.researchgate.net/publication/379536274_Polyp-SAM_transfer_SAM_for_polyp_segmentation
 33. Multi-Scale Efficient Transformer Attention Unet for Fast and High,
https://www.researchgate.net/publication/372320628_META-Unet_Multi-Scale_Efficient_Transformer_Attention_Unet_for_Fast_and_High-Accuracy_Polyp_Segmentation
 34. Endoscopy_old | HTIC IITMadras, <https://www.hticiitm.org/endoscopy-old>
 35. Healthcare Technology Innovation Centre(HTIC) - IIT Madras,
<https://www.iitm.ac.in/research/institute-research-centres/healthcare-technology-innovation-centre>
 36. FPGA Design Engineer (2 3 years) Medical & Video Systems - BeBee,
https://bebee.com/in/jobs/fpga-design-engineer-2-3-years-medical-video-systems-healthcare-technology-innovation-centre-chennai--techmap_in_59175599
 37. Complete List of Fellows - INSA,
https://insajournal.in/intranetinsa/completelist_fellow.php
 38. SRM Institute of Science and Technology (SRM University),
<https://collegedunia.com/articles/c-25896-happenings--srm-institute-of-science-and-technology-srm-university-chennai>
 39. Dr. Akella S Narasimha Raju - SRMIST,
<https://www.srmist.edu.in/faculty/dr-akella-s-narasimha-rajul>
 40. ColoRectalCADx: Expeditious Recognition of Colorectal Cancer,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9671728/>